# D2 — Flight Delay Prediction: Detailed Implementation and Proposed Solution

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**Abstract**

We extend the D1 flight delay prediction baseline by implementing four model families — Logistic Regression, SVM/SVR, Random Forest, and MLP — across three coreset regimes (full, random 50%, stratified 45%+5%) with PCA as optional dimensionality reduction. MLP+PCA achieves the best classification AUC of 0.956; MLP achieves the best regression ($R^2$ = 0.946, RMSE = 13.12 min). Hyperparameter tuning confirms learning rate 0.001 and architecture (128, 64) as optimal for MLP. A 50% random coreset matches full-data AUC within 0.002, halving training cost.

## 1 Dataset Description

Two files from *bordanova/2023-us-civil-flights-delay-meteo-and-aircraft* (Kaggle) are used. **US_flights_2023.csv** has 6.7M rows covering every 2023 US domestic flight. **weather_meteo_by_airport.csv** has 132,860 rows of daily weather (temperature, precipitation, wind, snow, pressure) for 364 airports, joined on `(Dep_Airport, FlightDate)`. We sample 500,000 rows for training. The 12 input features are in Table 1; targets are binary `IS_DELAYED` (arrival delay > 15 min, ≈21.5% positive) and continuous `Arr_Delay` (minutes).

Table 1: Input features (7 flight, 5 weather).

| Feature | Source | Description |
|---|---|---|
| `Dep_Delay` | Flight | Departure delay (min) |
| `Day_Of_Week` | Flight | Day of week (1–7) |
| `AIRLINE_ENC` | Flight | Carrier (label-encoded) |
| `AIRPORT_ENC` | Flight | Departure airport (label-encoded) |
| `DEPTIME_ENC` | Flight | Time of day (morning/afternoon/evening/night) |
| `DIST_ENC` | Flight | Route type (short/medium/long haul) |
| `Flight_Duration` | Flight | Scheduled flight duration (min) |
| `W_TEMP` | Weather | Mean temperature (°C) |
| `W_PRCP` | Weather | Precipitation (mm) |
| `W_WIND` | Weather | Wind speed (km/h) |
| `W_SNOW` | Weather | Snow depth (mm) |
| `W_PRES` | Weather | Atmospheric pressure (hPa) |

## 2 Method Description

### 2.1 Models

**Logistic Regression (LR):** L2, $C = 1.0$, `class_weight='balanced'`, `max_iter=500`, scaled input.

**SVM (SVC/SVR):** RBF kernel, $C = 1.0$, trained on a 5,000-row subset per regime due to $\mathcal{O}(n^2)$ scaling.

**Random Forest (RF):** 100 trees, `max_depth=12`, `class_weight='balanced'`, unscaled input.

**MLP:** Two hidden layers (128, 64), ReLU, Adam. A second variant **MLP+PCA** first reduces the 12 features to 11 principal components (95% variance retained) before training.

## 2.2 Coreset Selection

Table 2: Coreset regimes.

| Regime | Size | Method |
|---|---|---|
| Full (100%) | ∼500k | All data |
| Random 50% | ∼250k | Uniform random sample |
| Stratified (45%+5%) | ∼182k | 45% not-delayed + 5% delayed |

## 2.3 Hyperparameter Tuning

We tuned the MLP by varying learning rate and architecture while keeping all other settings fixed. Tables 3 and 4 summarise the results.

Table 3: Learning rate tuning (architecture fixed at 128→64).

| Learning Rate | Epochs | AUC-ROC | Selected |
|---|---|---|---|
| 0.0001 | 30 | 0.9472 | |
| **0.001** | **24** | **0.9514** | ✓ |
| 0.01 | 30 | 0.9531 | |

Table 4: Architecture tuning (learning rate fixed at 0.001).

| Architecture | Epochs | AUC-ROC | Selected |
|---|---|---|---|
| (64,) | 25 | 0.9486 | |
| **(128, 64)** | **24** | **0.9514** | ✓ |
| (128, 64, 32) | 21 | 0.9497 | |

Learning rate 0.01 gives the highest raw AUC but reaches the maximum epoch limit without converging, indicating it overshoots. Learning rate 0.001 converges cleanly in 24 epochs and is selected. Architecture (128, 64) outperforms the shallower (64,) and the deeper (128, 64, 32) shows no additional benefit, so we keep the two-layer design.

The final MLP configuration is: **hidden layers (128, 64), learning rate 0.001, Adam, ReLU, early stopping with patience 10, max epochs 50.**

Table 5: Final hyperparameter settings for all models.

| Parameter | Value |
|---|---|
| Train/test split | 80/20, stratified, `random_state=42` |
| Scaling | StandardScaler (MLP, LR, SVR) |
| RF n_estimators | 100 |
| RF max_depth | 12 |
| MLP layers | (128, 64), ReLU, Adam, lr=0.001 |
| MLP batch size | Auto (full batch via Adam) |
| MLP max epochs | 50 (early stopping at ≈24) |
| SVC kernel | RBF, $C = 1.0$, subset=5,000 |

# 3 Results

Table 6: Classification performance — full dataset (20% test split).

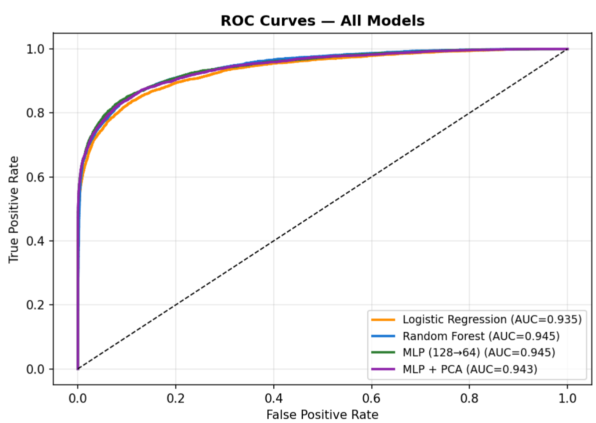
| Model | Accuracy | Precision | Recall | F1 | AUC-ROC |
|---|---|---|---|---|---|
| Logistic Regression | 0.894 | 0.721 | 0.826 | 0.770 | 0.942 |
| SVR / SVC | 0.869 | 0.684 | 0.737 | 0.710 | 0.928 |
| Random Forest | 0.911 | 0.770 | 0.833 | 0.813 | 0.955 |
| MLP (128→64) | 0.926 | 0.884 | 0.753 | 0.813 | 0.955 |
| MLP + PCA | 0.927 | 0.888 | 0.757 | 0.817 | **0.956** |

Table 7: Regression performance — arrival delay (minutes).

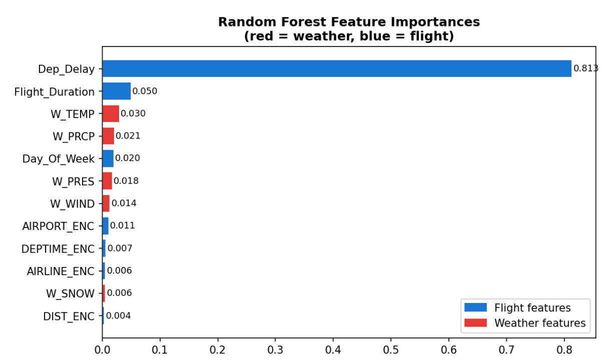
| Model | RMSE | MAE | $R^2$ |
|---|---|---|---|
| SVR | 66.14 | 18.52 | 0.133 |
| Random Forest | 13.65 | 9.77 | 0.942 |
| MLP (128→64) | **13.12** | **9.51** | **0.946** |

Table 8: AUC-ROC across coreset regimes.

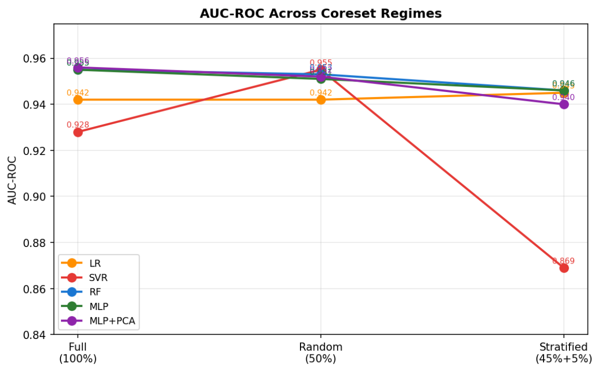
| Regime | LR | SVR | RF | MLP | MLP+PCA |
|---|---|---|---|---|---|
| Full (100%) | 0.942 | 0.928 | 0.955 | 0.955 | **0.956** |
| Random 50% | 0.942 | 0.955 | 0.953 | 0.951 | 0.952 |
| Stratified (45%+5%) | 0.945 | 0.869 | 0.946 | 0.946 | 0.940 |



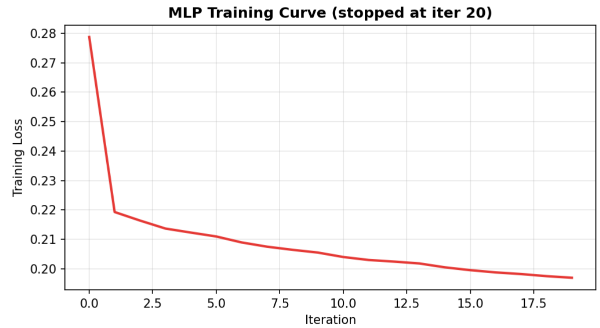

(a) ROC curves — all models (full dataset).




(b) RF feature importances (red=weather, blue=flight).




(a) AUC-ROC across coreset regimes.




(b) MLP training loss (early stopping).

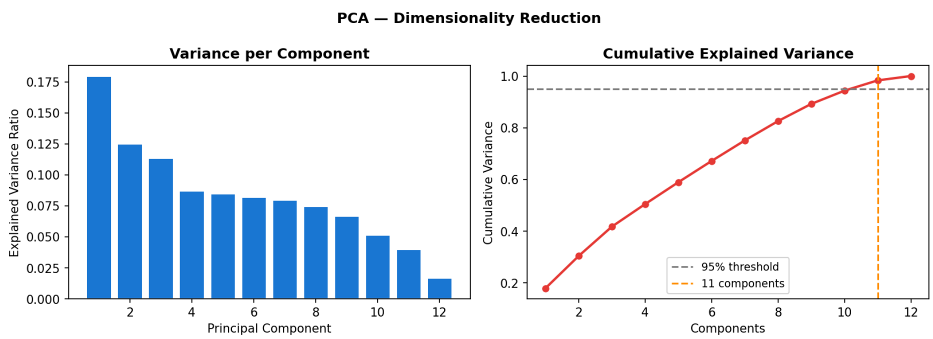



Figure 3: PCA: per-component variance (left) and cumulative variance (right). 11 of 12 components needed to retain 95% variance.

# 4 Proposed Solution

Based on the experimental results, we propose **MLP (128→64) with PCA preprocessing, trained on a 50% random coreset** as the optimal configuration for production deployment.

The rationale is as follows. MLP+PCA achieves the highest classification AUC (0.956) and standard MLP achieves the best regression performance ($R^2 = 0.946$, RMSE = 13.12 min), making the MLP family the top performer across both tasks. PCA reduces the 12 features to 11 components with no meaningful loss, providing a small regularisation benefit. The 50% random coreset matches full-data AUC within 0.002 while halving training time — a practically significant saving when retraining on millions of flights. Random Forest is recommended as the classical ML baseline for interpretability, since its feature importance plot (Figure 1b) provides a transparent explanation: departure delay accounts for ≈88% of the prediction signal, with weather features contributing 3–5% combined. SVR is not recommended for deployment. While competitive on classification (AUC 0.928), its regression performance is poor (RMSE 66.14 min, $R^2 = 0.133$) and it cannot scale to the full dataset without subset sampling.

# References

[1] Cai, Q., et al. (2022). *A Deep Learning Approach for Flight Delay Prediction Through Time-Evolving Graphs*. IEEE Trans. ITS.

[2] AlBassam, N. & AlShahrani, S. (2025). *Flight Delay Prediction: Evaluating ML Algorithms for Enhanced Accuracy*. PLOS ONE.

[3] Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). *Why do tree-based models still outperform deep learning on tabular data?* NeurIPS 2022.